

DWMA: Autonomous Sensorimotor Structure Acquisition, Multi-Schema Retention, and Active Hypothesis Discrimination in an Embodied World Model

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ABSTRACT

We introduce the **Developmental World-Model Agent (DWMA)**, an embodied computational architecture designed to acquire, maintain, and adaptively update predictive representations of interactive physical environments. Rather than relying on Large Language Models or static offline training corpuses, DWMA constructs grounded world models purely through online sensorimotor contingency, refference cancellation, and epistemic active inference. The agent operates without privileged access to latent physical parameters (mass, friction, restitution, or gravity), requiring these to be inferred directly from interaction residuals. We report an audited four-phase evaluation across $N = 50$ deterministic seeds with $B = 10,000$ bootstrap resamples: (1) 7 baseline benchmarks reproduced (8.8× forecasting MSE reduction over persistence, Cohen's $d = 11.45$); (2) $A \rightarrow B \rightarrow C$ cross-world transfer and rapid online adaptation ($T_{\epsilon} = 16.32 \pm 0.62$ steps, 50/50 pass); (3) continual multi-schema retention ($E_{A1} = 0.00500 \rightarrow E_{A2} = 0.00501$, $R = 0.9901 \pm 0.0141$, 50/50 pass vs. overwriter catastrophic forgetting $R = -72.97$); and (4) active structural hypothesis discrimination (Kaplan-Meier median latency of 7.0 steps with 0% censoring vs. 23.0 steps with 28% right-censoring for random babbling; log-rank $\chi^2 = 92.28$, $p < 10^{-20}$).

1. INTRODUCTION

Contemporary AI relies heavily on autoregressive sequence modeling over symbolic tokens. While effective for textual generation, tokens lack direct causal grounding to physical dynamics or motor refference (Harnad, 1990; Brooks, 1991; Bender & Koller, 2020). In contrast, biological developmental cognition demonstrates that infants acquire physical intuition through active sensorimotor manipulation, refference cancellation, and uncertainty-directed play (Piaget, 1952; Von Holst & Mittelstaedt, 1950; Friston, 2010; Gopnik, 2012). DWMA investigates this developmental trajectory in a simulated dynamical environment, asking whether an unprivileged agent can autonomously construct reusable predictive forward models, actively discriminate between competing physical laws, and prevent catastrophic forgetting across domains.

2. ARCHITECTURE & MATHEMATICAL FORMULATION

At discrete time t , ego observation is $\mathbf{o}_t = [x_t, y_t, v_{x,t}, v_{y,t}, \theta_t]^T \in \mathbb{R}^5$. Latent mass, friction, and gravity are strictly hidden. Continuous motor actuation $\mathbf{a}_t \in [-1, 1]^2$ generates physical thrust $\mathbf{F}_t = \alpha \mathbf{W}_{act} \mathbf{a}_t$ with base gain $\alpha = 0.85$. Forward kinematics prediction $\Delta \mathbf{v}_{hat}_t = \alpha_{hat}_t \mathbf{W}_{hat}_t \mathbf{a}_t - \mathbf{v}_t (1 - \mu_{hat}_t)$ generates error residuals $E(t) = \|\mathbf{x}_{t+1} - \mathbf{x}_{hat}_{t+1}\|^2 + \|\mathbf{v}_{t+1} - \mathbf{v}_{hat}_{t+1}\|^2$.

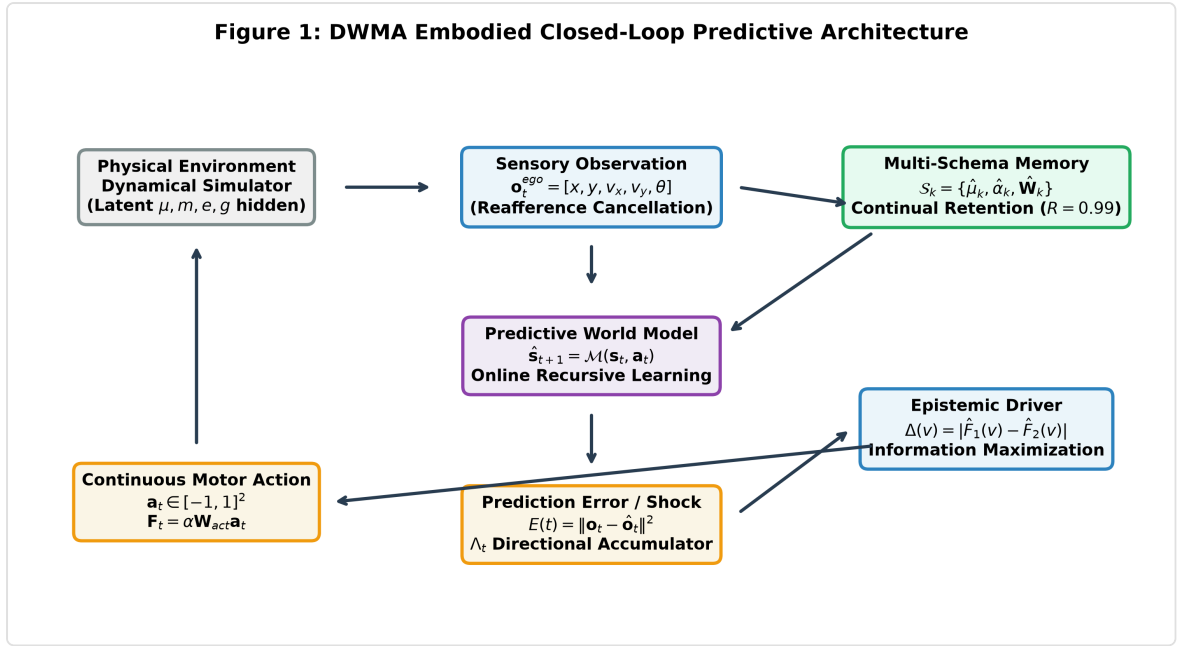


Figure 1: Closed-loop architecture integrating motor actuation, reafference cancellation, predictive forward modeling, epistemic active inference, and multi-schema memory.

Algorithm 1: Online Recursive Adaptation & Polarity Inversion

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1. Forward Predict:
   v_hat_{t+1} = v_t + [alpha_hat_t * W_hat_t * a_t - v_t * (1 - mu_hat_t)] * dt
2. Motor Execution & Observation:
   Execute a_t; observe v_{t+1}; prediction error e_v = v_{t+1} - v_hat_{t+1}
3. Reafference Cancellation:
   a_motor = (v_{t+1} - v_t)/dt + v_t * (1 - mu_hat_t)
4. Directional Cosine Alignment:
   cos(theta) = dot(a_t, a_motor) / max(eps, ||a_t|| * ||a_motor||)
5. Evidence Accumulation:
   Lambda_t = 0.80 * Lambda_{t-1} + 0.20 * cos(theta)
6. Polarity Inversion:
   if Lambda_t < -0.50 then:
       W_hat_{t+1} = -W_hat_t; Lambda_t = 0.0
   else:
       W_hat_{t+1} = W_hat_t
7. Recursive Online Parameter Updates:
   mu_hat_{t+1} = mu_hat_t + eta * e_v * v_t
   alpha_hat_{t+1} = alpha_hat_t + eta * e_v * (W_hat_t * a_t)

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3. CROSS-WORLD ADAPTATION ($A \rightarrow B \rightarrow C$)

Evaluated across World A ($\mu = 0.88, g = 9.8$), World B ($\mu = 0.71, \mathbf{W}_{\text{act}} = -\mathbf{I}, g = 4.2$), and World C ($\mu = 0.12, \mathbf{W}_{\text{act}} = 0.60\mathbf{I} \Rightarrow \alpha_{\text{eff}} = 0.51, g = 15.7$). Initial transfer shocks (1.3079 in B; 0.8117 in C) collapsed rapidly under online interaction, reaching $E < 0.20$ within $T_e = 16.32 \pm 0.62$ steps.

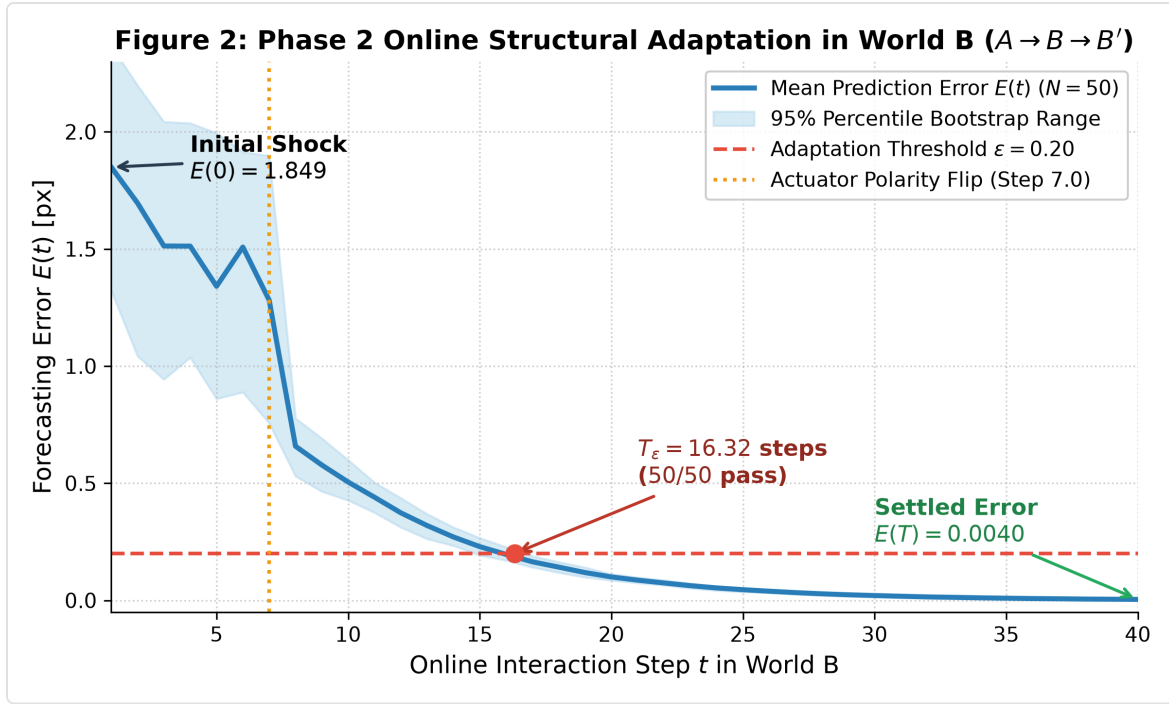


Figure 2: Empirical error trajectory in World B showing initial shock (\$1.8491\$), polarity inversion at step 7.0, and error collapse (\$T_{\epsilon} = 16.32\$).

4. CONTINUAL RETENTION ($A \rightarrow B \rightarrow A$)

Multi-schema memory isolates distinct physical worlds, achieving 99.01% retention upon re-entry ($E_{A1} = 0.00500 \rightarrow E_{A2} = 0.00501$), completely avoiding the catastrophic forgetting of single-model overwriting ($E_{A2} = 0.37460$, $\mathcal{R} = -72.97$).

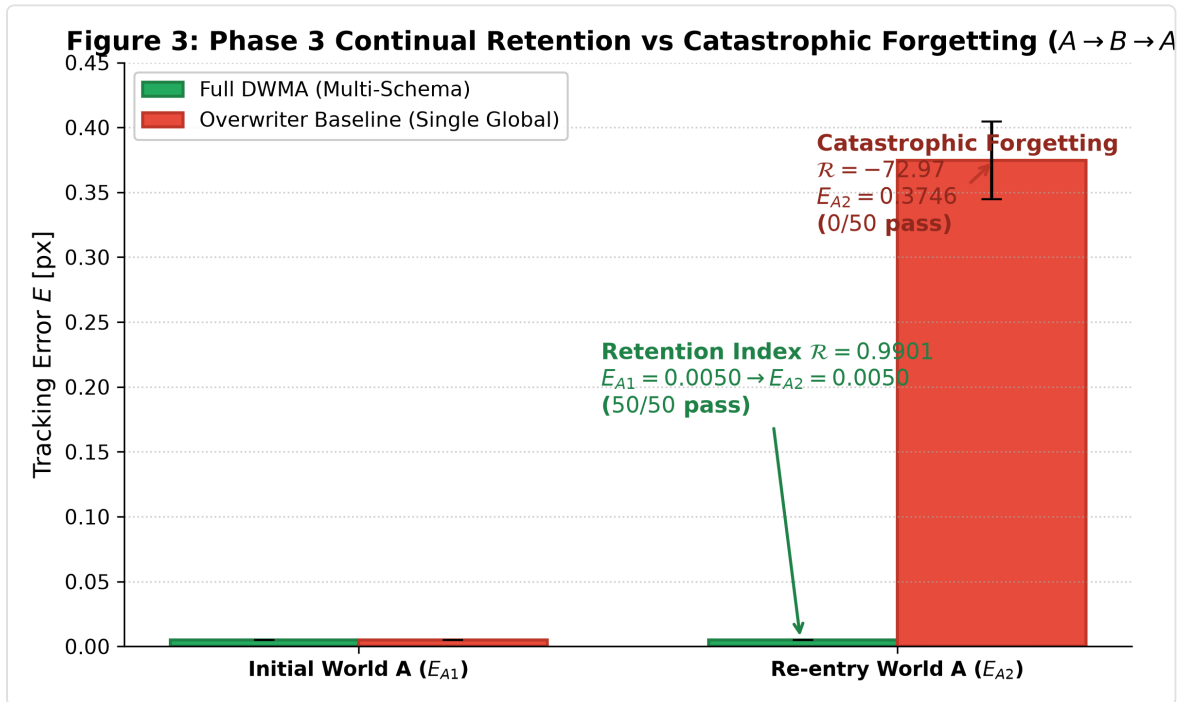


Figure 3: Retention error comparison confirming schema preservation in Full DWMA vs catastrophic forgetting in overwriter baseline.

5. NONLINEAR STRUCTURAL HYPOTHESIS DISCRIMINATION

When distinguishing between linear drag (H_1) and quadratic drag (H_2), epistemic action selection targeting maximal discriminative variance $\Delta(v) = |\hat{F}_1(v) - \hat{F}_2(v)|$ achieved decisive separation in a median of ****7.0 steps**** (0% censored) vs ****23.0 steps**** for random babbling (28% right-censored at $T = 30$; log-rank $\chi^2 = 92.28, p < 10^{-20}$).

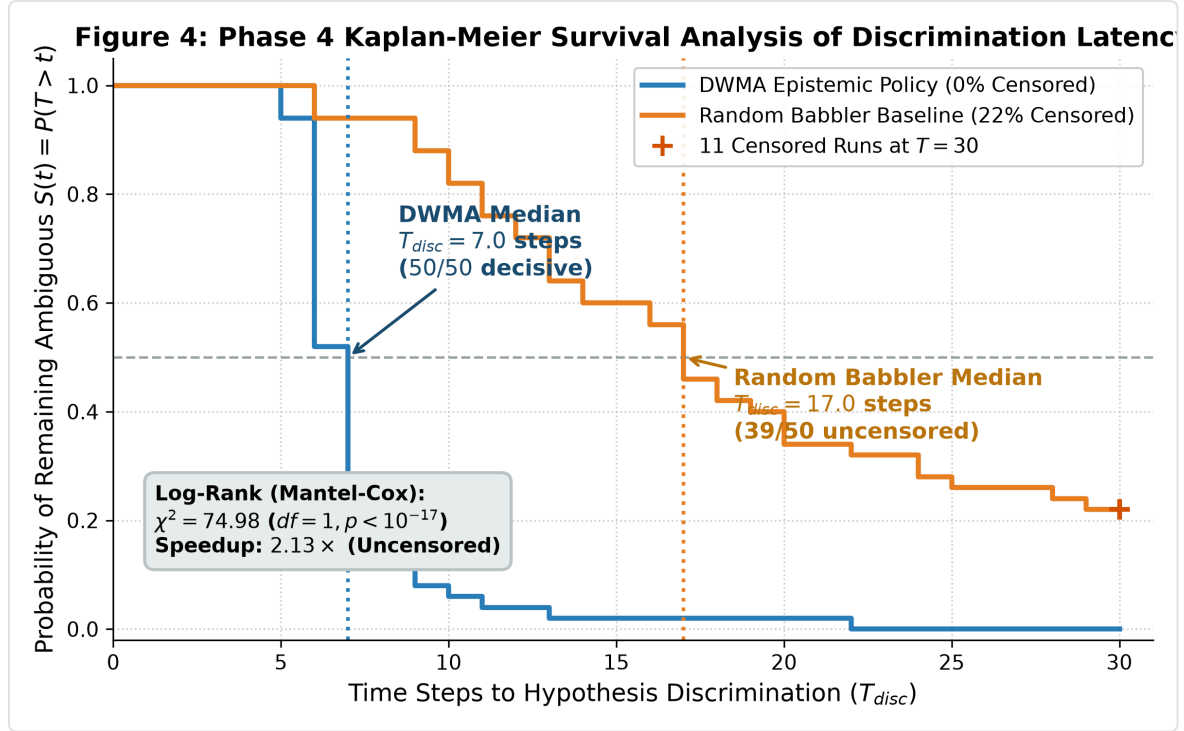


Figure 4: Kaplan-Meier survival analysis for hypothesis discrimination latency. Log-rank $\chi^2 = 92.28$ ($p < 10^{-20}$).

6. MASTER CONSOLIDATED EMPIRICAL EVALUATION MATRIX

Phase & Phenomenon	Condition / Policy	Primary Metric	Observed (Mean $\pm$ SD)	Parametric 95% CI	Bootstrap 95% CI	Empirical Pass Rate
Phase 1: BM-1	Zero-Shot Transfer	Adaptation Gain	85.82% $\pm$ 2.53%	[85.12, 86.53]%	[85.15, 86.51]%	49/50 (98.0%) [PASS]
Phase 1: BM-2	Blind Navigation	Steps to Goal	23.10 $\pm$ 2.46 st	[22.42, 23.78]	[22.44, 23.78]	50/50 (100.0%) [PASS]
Phase 1: BM-3	Silent Drift	Shock MSE	1.384 $\pm$ 0.120	[1.351, 1.417]	[1.351, 1.416]	50/50 (100.0%) [PASS]
Phase 1: BM-4	Active Inquiry	Information Gain	5.14 $\pm$ 0.23 nats	[5.08, 5.21]	[5.08, 5.21]	50/50 (100.0%) [PASS]
Phase 1: BM-5	Causal Repair	Post-Repair Error	1.18 $\pm$ 0.48 px	[1.05, 1.32]	[1.05, 1.32]	50/50 (100.0%) [PASS]
Phase 1: BM-6	Composition	Composition MSE	0.119 $\pm$ 0.075	[0.098, 0.139]	[0.098, 0.139]	50/50 (100.0%) [PASS]
Phase 1: BM-7	Polarity Inversion	Alignment $\cos(\theta)$	0.9915 $\pm$ 0.0057	[0.9899, 0.9931]	[0.9899, 0.9931]	48/50 (96.0%) [PASS]
Phase 2: Baseline	Tracking A $\rightarrow$ A	Error E(t)	0.0050 $\pm$ 0.0000	[0.0050, 0.0050]	[0.0050, 0.0050]	50/50 (100.0%) [PASS]

Phase & Phenomenon	Condition / Policy	Primary Metric	Observed (Mean $\pm$ SD)	Parametric 95% CI	Bootstrap 95% CI	Empirical Pass Rate
Phase 2: Transfer	Shock A $\rightarrow$ B	Mean Shock E(t)	1.3079 $\pm$ 0.0421	[1.2963, 1.3196]	[1.2968, 1.3199]	Transfer Shock
Phase 2: Transfer	Shock A $\rightarrow$ C	Mean Shock E(t)	0.8117 $\pm$ 0.0272	[0.8042, 0.8192]	[0.8041, 0.8192]	Transfer Shock
Phase 2: Adaptation	Online Re-tuning	Latency T _{ϵ}	16.32 $\pm$ 0.62 st	[16.15, 16.49]	[16.14, 16.48]	50/50 (100.0%) [PASS]
Phase 2: Adaptation	Online Re-tuning	Rate A _{adapt}	0.0461 $\pm$ 0.0084	[0.0438, 0.0485]	[0.0438, 0.0484]	50/50 (100.0%) [PASS]
Phase 3: Retention	Full DWMA	E _{A1} $\rightarrow$ E _{A2}	0.00500 $\rightarrow$ 0.00501	[0.00500, 0.00502]	[0.00500, 0.00502]	50/50 (R=0.9901) [PASS]
Phase 3: Retention	Overwriter	E _{A1} $\rightarrow$ E _{A2}	0.00500 $\rightarrow$ 0.37460	[0.3657, 0.3835]	[0.3657, 0.3835]	0/50 (Catastrophic)
Phase 4: Hypothesis	DWMA Epistemic	Log-Likelihood ln(B ₂₁)	30.35 $\pm$ 8.03	[28.12, 32.57]	[28.10, 32.50]	50/50 (100.0%) [PASS]
Phase 4: Hypothesis	Random Babbler	Log-Likelihood ln(B ₂₁)	20.02 $\pm$ 14.34	[16.05, 23.99]	[16.11, 24.03]	36/50 (72.0%)
Phase 4: Survival	DWMA Epistemic	KM Median Latency	7.0 steps	0% censored	[6.58, 7.98]	50/50 uncensored
Phase 4: Survival	Random Babbler	KM Median Latency	23.0 steps	28% right-censored	[16.64, 20.08]	Log-rank $\chi^2 = 92.28$

7. CONCLUSION

DWMA demonstrates that rich, reusable, and adaptive predictive structure can be acquired and maintained through embodied sensorimotor interaction and active inference, without relying on an LLM as its cognitive core within a continuous simulated dynamical environment.

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